

Spring Boot Web Project

Project Name: spring-learn

Technology Used: Spring Boot, Maven

Objective:

To create a Spring Boot web project using Maven, explore its structure, configure logging, and run the application successfully.

Steps Followed:

1. Project Initialization

- Used <https://start.spring.io> with the following configuration:
 - **Group:** com.cognizant
 - **Artifact:** spring-learn
 - **Dependencies:** Spring Web, Spring Boot DevTools
- Extracted the ZIP to Eclipse workspace.

2. Importing into Eclipse

- Followed:
 - File > Import > Maven > Existing Maven Projects
 - Selected the extracted folder and finished importing.

3. Building the Project

- Ran the Maven build command:

```
PS C:\Users\chait\eclipse-workspace\spring-learn> mvn clean package
```

```
arnApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ spring-learn ---
[INFO] Building jar: C:\Users\chait\eclipse-workspace\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:3.5.3:repackage (repackage) @ spring-learn ---
[INFO] Replacing main artifact C:\Users\chait\eclipse-workspace\spring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar with
repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\chait\eclipse-workspace\spring-learn\target\spring-learn-0.0.1
-SNAPSHOT.jar.original
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 5.992 s
[INFO] Finished at: 2025-07-10T11:58:19+05:30
[INFO]
PS C:\Users\chait\eclipse-workspace\spring-learn> |
```

4. Application Code Setup

- Located SpringLearnApplication.java inside:

- src/main/java/com/cognizant/spring_learn/

SpringLearningApplication.java:

```
package com.cognizant.spring_learn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import
org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    public static final Logger log =
LoggerFactory.getLogger(SpringLearnApplication.class);
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);

        log.info("Main Program Executed");
    }

}
```



```
:: Spring Boot ::
(v3.5.3)
```

(v3.5.3)

[	[	restartedMain]	c.c.s.p.r.i.n.g.l.e.a.r.n.S.p.r.i.n.g.L.e.a.r.n.A.p.p.l.i.c.a.t.i.o.n	: Starting SpringApplication using Java 21.0.7 with PID 560 (C:\Users\chaitvecl\p
[	[	restartedMain]	c.c.s.p.r.i.n.g.l.e.a.r.n.S.p.r.i.n.g.L.e.a.r.n.A.p.p.l.i.c.a.t.i.o.n	: No active profile set, falling back to 'default profile: 'default'
[	[	restartedMain]	e.DevTools.PropertyDefaultPostProcessor	: DevTools property defaults active! Set 'spring.devtools.add-property' to 'false' t
[	[	restartedMain]	e.DevTools.PropertyDefaultPostProcessor	: For additional web related logging consider setting the 'logging.level.web' property
[	[	restartedMain]	o.s.b.w.e.m.b.e.d.d.e.m.o.c.a.t.a.l.i.n.a.c.o.r.e.T.o.m.c.a.t.k.e.b.S.e.r.v.e.r	: Tomcat initialized with port 8080 (http)
[	[	restartedMain]	o.s.a.p.a.c.h.e.c.a.t.a.l.i.n.a.c.o.r.e.S.t.a.n.d.a.r.d.S.e.r.v.i.c.e	: Starting service [Tomcat]
[	[	restartedMain]	o.a.p.a.c.h.e.c.a.t.a.l.i.n.a.c.o.r.e.S.t.a.n.d.a.r.d.E.n.g.i.n.e	: Starting Servlet engine: [Apache Tomcat/10.1.42]
[	[	restartedMain]	o.a.c.c.c.[Tomcat].[localhost].[/]	: Initializing Spring embedded WebApplicationContext
[	[	restartedMain]	w.s.c.S.e.r.v.l.e.t.W.e.b.S.e.r.v.e.r.A.p.p.l.i.c.a.t.i.o.n.C.o.n.t.e.x.t	: Root WebApplicationContext: initialization completed in 1054 ms
[	[	restartedMain]	o.s.b.d.a.O.p.t.i.o.n.a.l.L.i.v.e.R.e.l.o.a.d.S.e.r.v.e.r	: LiveReload server is running on port 35729
[	[	restartedMain]	o.s.b.w.e.m.b.e.d.d.e.m.o.c.a.t.a.l.i.n.a.c.o.r.e.T.o.m.c.a.t.k.e.b.S.e.r.v.e.r	: Tomcat started on port 8080 (http) with context path '/'
[	[	restartedMain]	c.c.s.p.r.i.n.g.l.e.a.r.n.S.p.r.i.n.g.L.e.a.r.n.A.p.p.l.i.c.a.t.i.o.n	: Started SpringApplication in 1.972 seconds (process running for 2.422)
[	[	restartedMain]	c.c.s.p.r.i.n.g.l.e.a.r.n.S.p.r.i.n.g.L.e.a.r.n.A.p.p.l.i.c.a.t.i.o.n	: Main Program Executed

5. CONSOLE :

📁 Project Structure Overview:

Folder / File	Purpose
src/main/java	Application source code
src/main/resources	Configuration files (e.g., application.properties)
src/test/java	Placeholder for test classes
pom.xml	Maven build configuration

🔍 Maven Dependency Hierarchy:

```
PS C:\Users\chait\eclipse-workspace\spring-learn> mvn dependency:tree
```

```
[INFO] com.conginzant:spring-learn:jar:0.0.1-SNAPSHOT
[INFO] +- org.springframework.boot:spring-boot-starter-web:jar:3.5.3:compile
[INFO] | +- org.springframework.boot:spring-boot-starter:jar:3.5.3:compile
[INFO] | | +- org.springframework.boot:spring-boot-starter-logging:jar:3.5.3:compile
[INFO] | | | +- ch.qos.logback:logback-classic:jar:1.5.18:compile
[INFO] | | | | \- ch.qos.logback:logback-core:jar:1.5.18:compile
[INFO] | | +- org.apache.logging.log4j:log4j-to-slf4j:jar:2.24.3:compile
[INFO] | | | \- org.apache.logging.log4j:log4j-api:jar:2.24.3:compile
[INFO] | | \- org.slf4j:jul-to-slf4j:jar:2.0.17:compile
[INFO] | +- jakarta.annotation:jakarta.annotation-api:jar:2.1.1:compile
[INFO] | \- org.yaml:snakeyaml:jar:2.4:compile
[INFO] +- org.springframework.boot:spring-boot-starter-json:jar:3.5.3:compile
[INFO] | +- com.fasterxml.jackson.core:jackson-databind:jar:2.19.1:compile
[INFO] | | +- com.fasterxml.jackson.core:jackson-annotations:jar:2.19.1:compile
[INFO] | | \- com.fasterxml.jackson.core:jackson-core:jar:2.19.1:compile
[INFO] | +- com.fasterxml.jackson.datatype:jackson-datatype-jdk8:jar:2.19.1:compile
[INFO] | +- com.fasterxml.jackson.datatype:jackson-datatype-jsr310:jar:2.19.1:compile
[INFO] | \- com.fasterxml.jackson.module:jackson-module-parameter-names:jar:2.19.1:compile
[INFO] +- org.springframework.boot:spring-boot-starter-tomcat:jar:3.5.3:compile
[INFO] | +- org.apache.tomcat.embed:tomcat-embed-core:jar:10.1.42:compile
[INFO] | +- org.apache.tomcat.embed:tomcat-embed-el:jar:10.1.42:compile
[INFO] | \- org.apache.tomcat.embed:tomcat-embed-websocket:jar:10.1.42:compile
[INFO] +- org.springframework:spring-web:jar:6.2.8:compile
[INFO] | +- org.springframework:spring-beans:jar:6.2.8:compile
[INFO] | | \- io.micrometer:micrometer-observation:jar:1.15.1:compile
[INFO] | | \- io.micrometer:micrometer-commons:jar:1.15.1:compile
[INFO] | \- org.springframework:spring-webmvc:jar:6.2.8:compile
[INFO] +- org.springframework:spring-aop:jar:6.2.8:compile
[INFO] | +- org.springframework:spring-context:jar:6.2.8:compile
[INFO] | \- org.springframework:spring-expression:jar:6.2.8:compile
[INFO] +- org.springframework.boot:spring-boot-devtools:jar:3.5.3:runtime (optional)
[INFO] | +- org.springframework.boot:spring-boot:jar:3.5.3:compile
[INFO] | \- org.springframework.boot:spring-boot-autoconfigure:jar:3.5.3:compile
[INFO] \- org.springframework.boot:spring-boot-starter-test:jar:3.5.3:test
[INFO] | +- org.springframework.boot:spring-boot-test:jar:3.5.3:test
[INFO] | +- org.springframework.boot:spring-boot-test-autoconfigure:jar:3.5.3:test
[INFO] | +- com.jayway.jsonpath:json-path:jar:2.9.0:test
```

```
[INFO] | \- org.slf4j:slf4j-api:jar:2.0.17:compile
[INFO] +- jakarta.xml.bind:jakarta.xml.bind-api:jar:4.0.2:test
[INFO] | \- jakarta.activation:jakarta.activation-api:jar:2.1.3:test
[INFO] +- net.minidev:json-smart:jar:2.5.2:test
[INFO] | \- net.minidev:accessors-smart:jar:2.5.2:test
[INFO] |   \- org.ow2.asm:asm:jar:9.7.1:test
[INFO] +- org.assertj:assertj-core:jar:3.27.3:test
[INFO] | \- net.bytebuddy:byte-buddy:jar:1.17.6:test
[INFO] +- org.awaitility:awaitility:jar:4.2.2:test
[INFO] +- org.hamcrest:hamcrest:jar:3.0:test
[INFO] +- org.junit.jupiter:junit-jupiter:jar:5.12.2:test
[INFO] | +- org.junit.jupiter:junit-jupiter-api:jar:5.12.2:test
[INFO] | | +- org.opentest4j:opentest4j:jar:1.3.0:test
[INFO] | | +- org.junit.platform:junit-platform-commons:jar:1.12.2:test
[INFO] | | | \- org.apiguardian:apiguardian-api:jar:1.1.2:test
[INFO] | | +- org.junit.jupiter:junit-jupiter-params:jar:5.12.2:test
[INFO] | | \- org.junit.jupiter:junit-jupiter-engine:jar:5.12.2:test
[INFO] | |   \- org.junit.platform:junit-platform-engine:jar:1.12.2:test
[INFO] +- org.mockito:mockito-core:jar:5.17.0:test
[INFO] | +- net.bytebuddy:byte-buddy-agent:jar:1.17.6:test
[INFO] | \- org.objenesis:objenesis:jar:3.3:test
[INFO] +- org.mockito:mockito-junit-jupiter:jar:5.17.0:test
[INFO] +- org.skyscreamer:jsonassert:jar:1.5.3:test
[INFO] | \- com.vaadin.external.google:android-json:jar:0.0.20131108.vaadin1:test
[INFO] +- org.springframework:spring-core:jar:6.2.8:compile
[INFO] | \- org.springframework:spring-jcl:jar:6.2.8:compile
[INFO] +- org.springframework:spring-test:jar:6.2.8:test
[INFO] \- org.xmlunit:xmlunit-core:jar:2.10.2:test
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 20.480 s
[INFO] Finished at: 2025-07-10T11:02:47+05:30
[INFO] -----
PS C:\Users\chait\eclipse-workspace\spring-learn> |
```